Fiscal Year:	FY 2010	**Task Last Updated:**	FY 11/10/2010
PI Name:	Stuster, Jack W. Ph.D.		
Project Title:	Behavioral Issues Associated with Long Duration Space Missions: Review of Astronaut Journals		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Behavior and performance		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **BHP**:Behavioral Health & Performance (archival in 2017)		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	jstuster@anacapasciences.com	**Fax:**	FY
PI Organization Type:	INDUSTRY	**Phone:**	805-680-1315
Organization Name:	Anacapa Sciences, Inc.		
PI Address 1:	1516 Marquard Terrace		
PI Address 2:			
PI Web Page:	http://www.anacapasciences.com		
City:	Santa Barbara	**State:**	CA
Zip Code:	93101-4967	**Congressional District:**	24
Comments:	New address per PI (12/2012); previous address--301 East Carrillo Street, Santa Barbara, CA		
Project Type:	Flight	**Solicitation / Funding Source:**	ILSRA 2001
Start Date:	03/01/2003	**End Date:**	04/30/2010
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Shea, Cami1e	**Contact Phone:**	281-244-2017
Contact Email:	shea@dsls.usra.edu		
Flight Program:	ISS		
Flight Assignment:	ISS -- Beginning with Expedition 8 NOTE: Received extension to 4/30/2010 (from 1/31/2010) per PI and JSC (3/2010) NOTE: Received extension to 1/31/2010 (from 8/31/2009) per PI (9/2009) In flight development phase (data collection has begun)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NCC9-171		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This project is a focused study of human factors, behavior, and performance issues during expeditions to the International Space Station (ISS). The objective of the study is to obtain behavioral and human factors data from a systematic analysis of astronaut journals maintained during ISS increments. The investigator has studied conditions analogous to space craft in previous projects, including a two-year study (conducted for NASA with the cooperation of ESA and the French Polar Institute) in which content analyses were performed of diaries maintained by the station leaders and physicians at French Antarctic and South Indian Ocean remote duty stations. That study provided the first rank-ordering of the behavioral issues associated with isolation and confinement, based on quantitative data. In the current project, ISS crew maintain personal journals for subsequent review and content analysis. ISS expeditions provide unique research opportunities; this study represents the next logical step in the research program. See also http://www.nasa.gov/
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Astronauts have reported that participating in the Journals Experiment has helped them adapt to the isolation, confinement, and other special conditions of the ISS.
Task Progress:	Personal journals that were produced by ten subjects during ISS expeditions were analyzed to obtain information concerning a wide range of behavioral and human factors issues. Astronauts wrote most about their work, followed by outside communications (with mission control, family, and friends), adjustment to the conditions, interactions with crew mates, recreation/leisure, equipment (installation, maintenance), events (launches, docking, hurricanes, etc.), organization/management, sleep, and food. The study found evidence of a decline in morale during the third quarters of the missions and identified key factors that contribute to sustained adjustment and optimal performance during long-duration space expeditions. Astronauts reported that they benefited personally from writing in their journals because it helped maintain perspective on their work and relations with others. Responses to questions asked before, during, and after the expeditions show that living and working onboard the ISS is not as difficult as the astronauts anticipate before starting their six-month tours of duty. Recommendations include application of study results and continuation of the experiment to obtain additional data as crew size increases and operations evolve. The results were presented in a final report that was published in July 2010 by NASA as NASA/TM-2010-216130.
Bibliography Type:	Description: (Last Updated: 11/13/2019)
NASA Technical Documents	Stuster J. "Behavioral Issues Associated with Long-Duration Space Expeditions: Review and Analysis of Astronaut Journals." Houston, TX : NASA Lyndon B. Johnson Space Center, 62 p., 2010. NASA TM-2010-216130. , Jul-2010